Stage 4: Check-your-understanding questions

Q1. Why can AIC not settle South Carolina, but the case split can?

Table 4 compares H1 (network only), H2 (response only), and H3 (both). Their aggregate negative-binomial fits have the following results:

Model	AIC	Delta AIC	Predicted community share
H1	148.4	1.7	100%
H2	146.7	0.0	49%
H3	148.0	1.3	98%

AIC = 2q - 2 log(L_hat), where q is the number of fitted parameters. Relative to the minimum 146.7, the AIC differences are 1.7, 0.0, and 1.3. All are below 2, so AIC offers no clear discrimination on the observed aggregate series. This does not establish that the explanations are equally true.

For each two-week period j, the likelihood receives only the predicted total:

$$\mu_j = [C1(t_j) - C1(t_{j-1})] + [C2(t_j) - C2(t_{j-1})] = \mu_{1j} + \mu_{2j}.$$

For example, predicted splits (10, 90) and (51, 49) both yield a total of 100; at equal dispersion they give the same likelihood contribution for the observed aggregate count. The likelihood never observes group membership.

Officials reported approximately 90% of cases in the community. H1 predicts 100% (10 percentage points higher), H3 predicts 98% (8 higher), but H2 predicts 49% (41 lower). H2 additionally needs transmission to fall by 95% starting January 29, at the fitted search boundary. Thus H2 has the best aggregate AIC but an implausible case allocation. The independently reported case split strongly disfavors H2, without proving that public-health responses had no effect.

Q2. Derive the elasticity of critical coverage with respect to theta.

Proposition 3(ii), in the small-community limit, gives:

$$c_2^* = [1 - 1/(R_0 \theta)] / \varepsilon_2 = 1/\varepsilon_2 - (1/(\varepsilon_2 R_0)) \theta^{-1}.$$

Differentiate holding R_0 and ε_2 constant:

$$\partial c_2^* / \partial \theta = -(1/(\varepsilon_2 R_0))[-\theta^{-2}] = 1/(\varepsilon_2 R_0 \theta^2).$$

The normalised elasticity is:

$$\Upsilon = (\partial c_2^*/\partial \theta)(\theta/c_2^*) = 1/(\varepsilon_2 R_0 \theta c_2^*).$$

Since $\varepsilon_2 c_2^* = (R_0 \theta - 1)/(R_0 \theta)$, substitution yields:

$$\Upsilon = 1/(R_0 \theta - 1).$$

R_0 and θ enter symmetrically through their product, so the approximate elasticity with respect to R_0 is the same. For South Carolina, $R_0 \theta = 2.83$, giving $1/(2.83 - 1) = 1/1.83 = 0.54645$, or approximately 0.55. Section 5.2(B) reports a computed elasticity near 0.52. The small difference is expected because the analytic expression takes the small-community limit, while the numerical sensitivity uses the full two-group model and central differences.

As R0 theta approaches 1 from above, the elasticity diverges to positive infinity, while c2* approaches zero. This means even small absolute threshold changes are very large relative to a near-zero coverage requirement. It does not mean the required coverage becomes infinite. Below threshold, additional positive critical coverage is unnecessary in the limiting model.

Q3. Why does a longer latent period give a larger Rv?

For exponentially distributed SEIR stages, R = (1 + r/sigma)(1 + r/gamma). Use baseline R = 3.00, sigma = 1/10 per day, and gamma = 1/8 per day:

> 3 = (1 + 10r)(1 + 8r) = 1 + 18r + 80r^2.

> 80r^2 + 18r - 2 = 0; r = [-18 + sqrt(18^2 + 640)]/160 = 0.08155218 per day.

The positive root corresponds to epidemic growth, with a doubling time ln(2)/r = approximately 8.50 days. Hold this inferred growth rate fixed and change only the latent period:

> 8 days: R = (1 + 8r)(1 + 8r) = 2.73048.

> 12 days: R = (1 + 12r)(1 + 8r) = 3.26952.

Latent period	SEIR plug-in R	Table 5 refit Rv	Absolute difference
8 days	2.7305	2.74	0.0095
10 days	3.0000	3.00	0.0000
12 days	3.2695	3.26	0.0095

The observed epidemic growth rate r constrains the rise of the case curve, but the inferred reproduction number also depends on how long a transmission generation takes. A longer latent period delays secondary cases; a larger reproduction number is then needed to reproduce the same calendar-time growth. The simple homogeneous plug-in calculation differs slightly from Table 5 because Table 5 reoptimises a two-group preferential-mixing model with susceptible depletion, rather than holding all other parameters fixed.

Source: Working draft, October 25, 2026; Section 4.1 (p. 10), Table 4 (p. 16), Section 5.2(B) and Table 5 (pp. 18-19).